

Shelf-Life Prediction Models — v4 Targeted Recalibration Results Report

CHEESE_SHELF_LIFE_TARGETED_V4_READY_TO_TRAIN.xlsx (35,650 rows) retrained from scratch, tested against the same unmodified CHEESE_319_REAL_EXTERNAL_STRESS_TEST_CASES.xlsx · 319 cases · 34 real papers (DOI-traced) · generated 2026-08-19 17:29 UTC

Executive summary — v4 vs. v3 vs. v1, on the identical 319-case test set

- V4 fixed the one specific problem flagged in the v3 report.** A_STRICT_REAL (n=27, the tier singled out for R²): R²=-1.42 (v3) → R²=0.34 (v4, XGBoost), or R²=0.49 with LightGBM (v4's new default best model) — essentially back to v1's original quality. The 2 feta rows that broke this tier in v3 are now predicted accurately again.
- But v4 is not a net improvement — it traded one problem for two others.** Pooled R² across all 319 rows fell: 0.445 (v3) → 0.107 (v4).
- Semi-hard cheese collapsed:** R² 0.558 (v3, genuinely good) → 0.018 (v4), bias +16.3d → +68.1d. Root-caused below (\$8.4) to the v4 training labels for processed cheese, kashar, and gouda being substantially *inflated* versus v3.
- C_SAFETY_EVENT_REAL got substantially worse:** R² -0.95 (v3) → -4.47 (v4) — same root cause, since most of these rows are processed-cheese pathogen-challenge cases.
- Hard cheese continued improving (R² -2.93 → -0.52), and lower-bound compliance held roughly flat (54% → 55%).

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1. Methodology

319 real-condition cases traced to 34 distinct published papers (source_rule_id holds the actual DOI/PubMed link for each). data_origin = external_literature_real_condition and training_weight = 0 for every row — entirely disjoint from the training set, never used in fitting. Each row's quality_flag prefix is a confidence/comparison-semantics tier:

TIER	N	MEANING
A_STRICT_REAL	27	Exact reported value from the paper. Highest confidence.
A_RANGE_REAL	4	Paper reported a range rather than a single value.
B_OVERALL_REAL	23	Overall / nominal shelf-life claim stated in the paper.
B_LOWER_BOUND_REAL	54	Paper reports “no spoilage by day X” — true value is at least X.
C_STUDY_HORIZON_REAL	159	Value is bounded by how long the study ran (looser lower bound).
C_SAFETY_EVENT_REAL	52	Value tied to a specific pathogen/safety endpoint, not general spoilage.

All 4 live production models (Random Forest, LightGBM, XGBoost, Explainable Boosting Machine) were evaluated identically via ModelService.predict_one() — the exact same code path the deployed app uses, with zero retraining. Metrics: MAE (mean absolute error), RMSE, R² (coefficient of determination against the reported/imputed true value), MAPE, bias (mean signed error — positive means overprediction), and median absolute error (robust to outliers).

This report compares three model generations against the identical, unmodified 319-row file: **v1** (original training data, backed up under model_versions/v1_restored_baseline_2026-08-18_141539/), **v3** (broad recalibration of hard/semi-hard labels, 48,240 rows, backed up under model_versions/v3_recalibrated_2026-08-19_124027/), and **v4** (current live models, a more targeted recalibration on 35,650 rows — ~26,050 rows retained from the original v1-era base plus ~9,600 rows newly added/recalibrated under synthetic_independent_literature_calibrated_v4 / synthetic_recalibrated_v4). All single-version sections below (3 onward) reflect

v4, the current live app; §2 puts all three side by side. XGBoost is used as the comparison model throughout for architecture-controlled consistency across versions, even though v4's own best-by-validation model is now LightGBM (noted where relevant).

2. v1 vs. v3 vs. v4 headline comparison (XGBoost, identical 319 rows)

Same evaluation script, same test file, unmodified, run three times against three different sets of trained artifacts. ▲/▼ reflect the v3→v4 change in R² beyond ±0.02; smaller shifts are marked ≈ flat.

SEGMENT	R ² (V1)	R ² (V3)	R ² (V4)	BIAS (V1)	BIAS (V3)	BIAS (V4)	V3→V4
Overall (pooled, all 319, all tiers)	-0.466	0.445	0.107	+19.5d	+3.0d	+16.8d	▼ regressed
A_STRICT_REAL (n=27)	0.487	-1.422	0.339	+0.0d	+1.2d	-0.7d	▲ improved
C_STUDY_HORIZON_REAL (n=159, largest tier)	-0.790	0.492	0.235	+36.8d	+11.9d	+36.3d	▼ regressed
C_SAFETY_EVENT_REAL (n=52)	-1.726	-0.946	-4.472	+26.6d	+21.1d	+36.9d	▼ regressed
Hard cheese (n=35)	-22.668	-2.929	-0.523	+150.5d	+38.3d	+16.6d	▲ improved
Semi-hard cheese (n=114)	0.148	0.558	0.018	+30.5d	+16.3d	+68.1d	▼ regressed
Soft cheese (n=170)	0.515	0.502	0.341	-14.8d	-13.2d	-17.5d	▼ regressed

Reading this table honestly: v4 is a genuine win on exactly one thing — A_STRICT_REAL, the tier this project has repeatedly been asked to report R² on. Everywhere else the picture is mixed-to-worse: hard cheese kept improving, but semi-hard cheese and C_SAFETY_EVENT_REAL both regressed sharply, and pooled R² fell back toward v1-era territory. This is not "v4 is better" or "v4 is worse" than v3 — it is a different, narrower trade-off, root-caused in §8.

3. Overall results — all 319 rows, all tiers pooled, every model (v4)

All models, ranked by MAE

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
Random Forest	319	40.39d	63.81d	0.179	212.7%	+16.90d	20.09d
XGBoost	319	43.94d	66.54d	0.107	219.8%	+16.85d	21.96d
LightGBM	319	44.04d	67.20d	0.089	218.9%	+17.54d	22.58d
Explainable Boosting Machine	319	47.03d	70.79d	-0.011	204.0%	+15.13d	25.00d

The current app-selected best model is LightGBM (lowest validation RMSE on the v4 split, a change from XGBoost in v1/v3) — not necessarily the best on this real test set; comparison tables in this report keep using XGBoost throughout for cross-version consistency.

4. Results by confidence tier — every model (v4)

R² on tiers with a narrow true-value range (e.g. A_RANGE_REAL, n=4) is statistically unstable by construction — small denominators make R² swing wildly. MAE/bias are more reliable for those tiers.

A_STRICT_REAL (n=27)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
Random Forest	27	2.42d	3.31d	0.314	28.3%	+0.00d	1.37d
LightGBM	27	1.93d	2.86d	0.488	20.8%	-0.67d	1.50d
XGBoost	27	2.34d	3.25d	0.339	22.9%	-0.70d	1.60d
Explainable Boosting Machine	27	7.51d	8.88d	-3.927	85.6%	-2.91d	8.00d

Exact reported value from the paper. Highest confidence.

A_RANGE_REAL (n=4)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
Random Forest	4	4.20d	4.26d	-0.721	120.7%	+2.58d	4.33d
LightGBM	4	5.99d	6.03d	-2.452	157.8%	+2.46d	5.90d

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
XGBoost	4	6.71d	6.75d	-3.322	175.9%	+2.71d	6.49d
Explainable Boosting Machine	4	3.45d	5.37d	-1.733	52.4%	-2.48d	1.52d

Paper reported a range rather than a single value.

B_OVERALL_REAL (n=23)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
Random Forest	23	8.59d	25.25d	0.878	28.6%	-6.08d	2.49d
LightGBM	23	8.95d	27.06d	0.860	26.2%	-6.85d	1.69d
XGBoost	23	10.19d	27.89d	0.851	33.9%	-9.39d	4.80d
Explainable Boosting Machine	23	9.18d	21.33d	0.913	47.6%	-5.79d	5.08d

Overall / nominal shelf-life claim stated in the paper.

B_LOWER_BOUND_REAL (n=54)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
Random Forest	54	45.56d	67.80d	-0.048	54.8%	-34.13d	28.16d
LightGBM	54	51.30d	73.84d	-0.242	62.1%	-40.37d	39.94d
XGBoost	54	50.77d	71.83d	-0.176	63.6%	-38.59d	38.88d
Explainable Boosting Machine	54	69.70d	90.33d	-0.859	109.7%	-45.39d	47.39d

Paper reports "no spoilage by day X" — true value is at least X.

C_STUDY_HORIZON_REAL (n=159)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
Random Forest	159	46.99d	62.82d	0.356	127.3%	+31.59d	27.24d
LightGBM	159	52.75d	69.17d	0.219	154.1%	+36.83d	39.60d
XGBoost	159	53.22d	68.47d	0.235	156.8%	+36.27d	37.97d
Explainable Boosting Machine	159	54.88d	72.86d	0.134	162.5%	+40.13d	30.15d

Value is bounded by how long the study ran (looser lower bound).

C_SAFETY_EVENT_REAL (n=52)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
Random Forest	52	51.43d	88.58d	-5.036	822.0%	+45.03d	17.01d
LightGBM	52	50.15d	84.17d	-4.449	773.1%	+40.11d	20.19d
XGBoost	52	47.87d	84.34d	-4.472	762.4%	+36.85d	15.21d
Explainable Boosting Machine	52	40.10d	76.08d	-3.452	571.0%	+21.55d	15.00d

Value tied to a specific pathogen/safety endpoint, not general spoilage.

5. A_STRICT_REAL deep dive (n=27) — the tier requested for R², and the one v4 fixed

These are the 27 rows where a paper reported an exact shelf-life value with no ambiguity, no bound, no imputation of the outcome itself. This is the fairest, highest-confidence test of the model's actual predictive accuracy.

A_STRICT_REAL, ranked by MAE (v4)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
LightGBM	27	1.93d	2.86d	0.488	20.8%	-0.67d	1.50d
XGBoost	27	2.34d	3.25d	0.339	22.9%	-0.70d	1.60d
Random Forest	27	2.42d	3.31d	0.314	28.3%	+0.00d	1.37d
Explainable Boosting Machine	27	7.51d	8.88d	-3.927	85.6%	-2.91d	8.00d

This tier recovered under v4 (R^2 -1.42 in v3 $\rightarrow$ 0.34 with XGBoost, 0.49 with LightGBM — both close to or above v1's original 0.487). Excluding the 2 feta rows that broke this tier in v3, the remaining 25 rows score MAE=2.28d, R^2 =0.28; the feta pair itself is now MAE=3.0d instead of the ~19.5d it was in v3. Full three-way trace in §8.3.

The Explainable Boosting Machine remains the weakest model on this tier in v4 (R^2 = -3.93) — worth knowing if EBM is being used anywhere for its explainability rather than its accuracy.

6. Results by cheese category — every model (v4)

Cheese category: soft (n=170)

MODEL / GROUP	N	MAE	RMSE	R^2	MAPE	BIAS	MEDIAN AE
Random Forest	170	18.85d	36.85d	0.406	42.6%	-17.03d	8.91d
LightGBM	170	20.28d	40.29d	0.290	44.5%	-18.16d	6.01d
XGBoost	170	20.55d	38.83d	0.341	48.3%	-17.47d	9.42d
Explainable Boosting Machine	170	28.11d	50.00d	-0.094	76.4%	-20.23d	14.00d

Cheese category: semi_hard (n=114)

MODEL / GROUP	N	MAE	RMSE	R^2	MAPE	BIAS	MEDIAN AE
Random Forest	114	75.01d	93.64d	0.090	492.4%	+65.82d	51.28d
LightGBM	114	81.92d	97.62d	0.011	504.3%	+69.19d	70.33d
XGBoost	114	81.69d	97.30d	0.018	505.6%	+68.10d	69.05d
Explainable Boosting Machine	114	79.22d	97.73d	0.009	415.8%	+68.34d	71.00d

Cheese category: hard (n=35)

MODEL / GROUP	N	MAE	RMSE	R^2	MAPE	BIAS	MEDIAN AE
Random Forest	35	32.25d	44.13d	-0.353	128.0%	+22.38d	29.83d
LightGBM	35	36.05d	47.29d	-0.553	137.0%	+22.75d	35.97d
XGBoost	35	34.65d	46.84d	-0.523	121.9%	+16.63d	30.33d
Explainable Boosting Machine	35	34.08d	49.16d	-0.678	133.7%	+13.58d	24.08d

7. Control (untreated) vs. treatment rows — every model (v4)

Control (untreated) (n=190)

MODEL / GROUP	N	MAE	RMSE	R^2	MAPE	BIAS	MEDIAN AE
Random Forest	190	36.73d	64.45d	0.422	264.8%	+8.80d	14.66d
LightGBM	190	37.45d	63.51d	0.439	254.0%	+5.32d	15.74d
XGBoost	190	37.43d	63.80d	0.434	252.9%	+4.05d	15.00d
Explainable Boosting Machine	190	36.57d	61.72d	0.470	213.4%	+0.51d	17.49d

Treatment (n=129)

MODEL / GROUP	N	MAE	RMSE	R^2	MAPE	BIAS	MEDIAN AE
Random Forest	129	45.79d	62.85d	-1.652	135.9%	+28.83d	27.07d
LightGBM	129	53.74d	72.29d	-2.509	167.3%	+35.55d	42.24d
XGBoost	129	53.54d	70.37d	-2.325	171.1%	+35.70d	45.27d
Explainable Boosting Machine	129	62.43d	82.34d	-3.552	190.1%	+36.67d	52.87d

8. Root cause analysis

8.1 — Extrapolation was checked and ruled out (again, on v4)

Same check as in the v1 and v3 reports: if the model were simply being asked to extrapolate past its training range, restricting to in-range rows only should improve results.

In-range vs. out-of-range inputs (XGBoost, v4)

MODEL / GROUP	N	MAE	RMSE	R ²	MAPE	BIAS	MEDIAN AE
In-range (n=308)	308	44.64d	67.16d	0.118	225.2%	+17.38d	22.56d
Out-of-range (n=11)	11	24.31d	45.74d	-28.578	70.3%	+2.11d	12.75d

Same result as before: in-range performs worse, not better (R^2 0.12 in-range vs -28.58 out-of-range). Extrapolation is still not the explanation for v4's weak spots — both the A_STRICT_REAL fix and the new semi-hard regression are in-distribution calibration effects.

8.2 — Hard cheese: the v1→v3→v4 label and prediction arc

Parmigiano Reggiano. Real case: 597 days already aged, paper reports **90 days**. Matching synthetic training rows (hard category, ripening within $\pm 30\%$ of the real case, controls only) — median labeled shelf life: v1 **242.7d** (n=95, inflation 2.70 $\times$) → v3 **121.3d** (n=311, inflation 1.35 $\times$) → v4 **115.6d** (n=205, inflation 1.28 $\times$). The label has kept moving toward the real value across all three versions.

The model's live output has partly caught up. Mean XGBoost prediction across the Parmigiano test rows: v3 **206.4d** → v4 **123.2d** — still well above the real value (90d), but the gap between label and live output that the v3 report flagged has narrowed, consistent with hard cheese's continued (if incomplete) improvement in \$2 and \$6.

Reading: hard cheese is the one category recalibrated consistently across both v3 and v4, and it shows a consistent, monotonic improvement in both the label and the live prediction. It is the model of what "targeted recalibration done right" looks like — contrast with \$8.4.

8.3 — A_STRICT_REAL / feta: broke in v3, fixed in v4

Row-by-row comparison of the 2 feta rows across all three versions:

ROW ID	TRUE (D)	V1 PRED	V1 ERR	V3 PRED	V3 ERR	V4 PRED	V4 ERR
EXT_REAL_0120	10.0	9.5	-0.5	25.0	+15.0	13.8	+3.8
EXT_REAL_0121	17.0	22.0	+5.0	41.0	+24.0	19.3	+2.3

v1 predicted both rows within about a day; v3 badly overpredicted both (this is what broke the tier's R^2); v4 is back to within a few days of the true value on both rows. Since this report still does not know *why* the v3 training data mislabeled feta specifically, it also cannot yet confirm whether v4's fix was a deliberate, targeted correction to that same root cause, or an incidental side effect of a broader adjustment to the soft-cheese portion of the label set. Either way, the outcome on these two specific rows is unambiguously better.

8.4 — NEW in v4: semi-hard cheese label inflation (processed cheese, kashar, gouda)

Unlike hard cheese, semi-hard cheese was not a target of the v3 recalibration and scored well in v3 ($R^2=0.56$). In v4 it collapsed ($R^2=0.02$, bias +68.1d). Direct comparison of the v3 vs. v4 training labels for the three semi-hard food matrices that dominate this test set's semi-hard rows:

FOOD MATRIX	N (V3)	MEDIAN LABEL (V3)	N (V4)	MEDIAN LABEL (V4)	CHANGE
processed cheese	2100	150.0d	1480	230.5d	+54%
kashar	2100	80.0d	1155	144.2d	+80%
gouda	2100	119.5d	1870	246.3d	+106%

Proven, not inferred: all three matrices' v4 training labels are substantially higher than their v3 labels — the opposite direction from what v3 did for hard cheese. This directly explains both the semi-hard collapse and most of the C_SAFETY_EVENT_REAL collapse, since the majority of C_SAFETY_EVENT_REAL rows in this test set are processed-cheese pathogen-challenge cases. Row-level confirmation for the processed-cheese C_SAFETY_EVENT_REAL rows:

ROW ID	TRUE (D)	STORAGE	INDICATOR	V3 PRED	V4 PRED
EXT_REAL_0224	270.0	4°C	listeria_monocytogenes	148.1	234.5
EXT_REAL_0225	14.0	5°C	listeria_monocytogenes	134.6	241.7
EXT_REAL_0226	14.0	5°C	salmonella	137.4	258.0
EXT_REAL_0227	14.0	5°C	staphylococcus_aureus	138.1	258.1
EXT_REAL_0228	3.0	5°C	pseudomonas_spp	131.9	250.0
EXT_REAL_0229	3.0	5°C	lactic_acid_bacteria	135.1	250.8
EXT_REAL_0232	14.0	22°C	listeria_monocytogenes	37.6	94.3
EXT_REAL_0233	14.0	22°C	salmonella	43.1	108.1
EXT_REAL_0234	14.0	22°C	staphylococcus_aureus	37.2	107.9
EXT_REAL_0235	3.0	22°C	pseudomonas_spp	37.4	103.3
EXT_REAL_0236	3.0	22°C	lactic_acid_bacteria	37.5	101.1
EXT_REAL_0300	4.0	30°C	listeria_monocytogenes	33.1	57.0
EXT_REAL_0301	4.0	30°C	staphylococcus_aureus	32.7	70.2
EXT_REAL_0302	4.0	30°C	salmonella	39.7	70.4
EXT_REAL_0303	4.0	30°C	escherichia_coli	32.7	70.2

The model is genuinely temperature-sensitive here (predictions drop as storage temperature rises across both versions) but is not sensitive to indicator_type/challenge context at all — a real background-safety control row (true 270d) and a deliberate pathogen-challenge row at the same temperature (true 3–14d) get nearly identical predictions in both v3 and v4. v4's inflated processed-cheese label made this pre-existing blind spot substantially worse in absolute terms, but did not create it — it was already present in v3 (see the v3-pred column above), just less severely.

9. Lower-bound compliance, v1 / v3 / v4

For tiers where the reported value is a floor (paper states "no spoilage observed by day X" — the true shelf life is at least X), a correct model should predict at or above that floor.

55% of 265 lower-bound-semantics predictions (XGBoost, v4) meet or exceed the reported floor (v1: 57%, v3: 54%). **120 rows under-predict** a duration the literature has already demonstrated is safely exceeded. Materially unchanged across all three versions — neither recalibration targeted bound semantics specifically.

10. Per-paper breakdown — all 34 papers (XGBoost, v4), ranked by MAE (worst first)

PAPER (DOI / LINK)	N	MAE	R ²	BIAS
https://pubmed.ncbi.nlm.nih.gov/18348389/	14	150.49	-24.333	+150.5
https://doi.org/10.1002/pts.2483	6	140.51	—	+140.5
https://pubmed.ncbi.nlm.nih.gov/12929846/	2	128.97	-665.356	+129.0
https://doi.org/10.1016/j.fm.2018.02.005	16	106.63	—	-106.6
https://doi.org/10.1016/j.fm.2010.04.017	2	82.95	-3.048	-83.0
https://doi.org/10.1016/j.idairyj.2023.105596	60	75.30	-42.277	+75.3
https://doi.org/10.1111/j.1365-2621.2010.02465.x	10	74.40	—	+74.4
https://doi.org/10.3168/jds.2024-25496	9	64.80	—	-64.8
https://doi.org/10.4315/0362-028X-61.3.290	4	62.95	—	+63.0
https://doi.org/10.3168/jds.2016-11658	6	52.79	-266.153	+52.8
https://doi.org/10.3168/jds.S0022-0302(93)77424-X	8	43.24	-4.353	-43.2
https://doi.org/10.1016/j.idairyj.2013.05.004	1	41.69	—	+41.7

PAPER (DOI / LINK)	N	MAE	R ²	BIAS
https://doi.org/10.1106/108201302028118	9	33.15	—	+33.2
https://pubmed.ncbi.nlm.nih.gov/9172439/	2	29.18	-1.512	-29.2
https://doi.org/10.3168/jds.2011-4219	12	23.63	-0.035	-23.6
https://doi.org/10.3168/jdsc.2022-0219	6	22.75	—	-22.8
https://doi.org/10.1016/j.ijfoodmicro.2021.109337	7	20.50	—	-20.5
https://doi.org/10.1021/jf902774z	9	19.88	—	-19.9
https://pubmed.ncbi.nlm.nih.gov/24031971/	2	17.27	—	-17.3
https://pubmed.ncbi.nlm.nih.gov/29477526/	1	16.61	—	-16.6
https://doi.org/10.1016/j.foodcont.2004.10.013	2	15.85	—	-15.9
https://pubmed.ncbi.nlm.nih.gov/29960774/	7	13.41	-1.413	-13.4
https://doi.org/10.1016/j.lwt.2013.04.007	18	12.30	—	+0.9
https://doi.org/10.3390/pathogens15010065	10	11.67	-1.470	+2.2
https://doi.org/10.1111/j.1365-2621.1994.tb14701.x	15	11.40	—	-11.4
https://doi.org/10.4315/0362-028X.JFP-14-047	16	8.88	—	-1.8
https://doi.org/10.1111/j.1365-2672.1991.tb02941.x	3	6.93	—	-6.9
https://pubmed.ncbi.nlm.nih.gov/25829584/	2	5.06	-0.042	-5.1
https://doi.org/10.1016/j.foohum.2023.09.010	12	4.78	-2.314	-4.8
https://pubmed.ncbi.nlm.nih.gov/27800345/	9	4.26	0.032	+0.6
https://doi.org/10.3168/jds.S0022-0302(92)78054-0	6	3.91	0.537	-1.8
https://pmc.ncbi.nlm.nih.gov/articles/PMC7270376/	16	2.24	0.597	+2.2
https://pubmed.ncbi.nlm.nih.gov/27070568/	1	1.73	—	+1.7
https://www.sciencedirect.com/science/article/pii/S0141813022008996	16	1.36	-0.050	-0.4

R^2 is undefined (—) for papers contributing a single true-value cluster with near-zero variance — a data artifact of R^2 's definition, not a model failure indicator for those rows. By worst MAE alone, <https://pubmed.ncbi.nlm.nih.gov/18348389/> ($n=14$) ranks first, but at only 4% of the test set it barely moves pooled statistics. The paper that actually drags down pooled numbers the most is the one combining a large sample with poor accuracy:

<https://doi.org/10.1016/j.idairyj.2023.105596> ($n=60$, ~19% of the whole test set, MAE=75.3d, $R^2=-42.28$).

11. Worst 30 individual predictions (XGBoost, v4, by absolute error)

ROW	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	PRED (D)	ERROR (D)
EXT_REAL_0227	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	258.1	+244.1
EXT_REAL_0226	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	258.0	+244.0
EXT_REAL_0229	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	3.0	250.8	+247.8
EXT_REAL_0228	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	3.0	250.0	+247.0
EXT_REAL_0225	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	241.7	+227.7
EXT_REAL_0231	processed cheese	semi_hard	C_STUDY_HORIZON_REAL	84.0	280.7	+196.7
EXT_REAL_0230	processed cheese	semi_hard	C_STUDY_HORIZON_REAL	84.0	258.1	+174.1
EXT_REAL_0279	feta	soft	B_LOWER_BOUND_REAL	180.0	25.6	-154.4
EXT_REAL_0278	feta	soft	B_LOWER_BOUND_REAL	180.0	27.3	-152.7
EXT_REAL_0280	feta	soft	B_LOWER_BOUND_REAL	180.0	29.2	-150.8
EXT_REAL_0131	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	184.6	+139.6
EXT_REAL_0277	feta	soft	B_LOWER_BOUND_REAL	180.0	37.3	-142.7
EXT_REAL_0130	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	181.6	+136.6
EXT_REAL_0128	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	181.5	+136.5

ROW	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	PRED (D)	ERROR (D)
EXT_REAL_0196	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	190.3	+145.3
EXT_REAL_0197	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	190.3	+145.3
EXT_REAL_0141	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	179.8	+134.8
EXT_REAL_0195	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	185.5	+140.5
EXT_REAL_0194	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	184.3	+139.3
EXT_REAL_0127	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	176.4	+131.4
EXT_REAL_0140	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	176.7	+131.7
EXT_REAL_0129	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	176.7	+131.7
EXT_REAL_0286	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	233.4	-131.6
EXT_REAL_0138	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	174.5	+129.5
EXT_REAL_0282	feta	soft	B_LOWER_BOUND_REAL	180.0	54.7	-125.3
EXT_REAL_0137	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	170.6	+125.6
EXT_REAL_0139	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	171.0	+126.0
EXT_REAL_0284	feta	soft	B_LOWER_BOUND_REAL	180.0	58.2	-121.8
EXT_REAL_0283	feta	soft	B_LOWER_BOUND_REAL	180.0	58.3	-121.7
EXT_REAL_0193	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	181.7	+136.7

12. Issues identified (v4 state)

- Semi-hard cheese collapsed, root-caused to label inflation.** R^2 0.56 (v3) $\rightarrow$ 0.02 (v4), bias +16.3d $\rightarrow$ +68.1d (n=114). Processed cheese/kashar/gouda training labels rose substantially from v3 to v4 (\$8.4) — the opposite direction of the intended recalibration.
- C_SAFETY_EVENT_REAL got substantially worse** (R^2 -0.95 $\rightarrow$ -4.47, n=52), same root cause as above, compounding a pre-existing blind spot: the model is not sensitive to pathogen-challenge context (indicator_type) at all, in v3 or v4.
- A_STRICT_REAL is fixed.** R^2 -1.42 (v3) $\rightarrow$ 0.34 (v4, XGBoost) / 0.49 (v4, LightGBM). The specific feta overprediction that broke this tier in v3 no longer reproduces. Root cause of why feta moved in v3 is still not identified, only that v4 no longer exhibits the symptom.
- Hard cheese kept improving, still imperfect.** R^2 -2.93 (v3) $\rightarrow$ -0.52 (v4), and the Parmigiano label-vs-live-output gap flagged in the v3 report narrowed (206d $\rightarrow$ 123d live output, vs a 116d v4 label and a 90d real value) but has not closed.
- Pooled/overall statistics fell back toward v1-era levels** (R^2 0.445 v3 $\rightarrow$ 0.107 v4) — driven almost entirely by the semi-hard/safety-event regression above, not a broad-based decline.
- Lower-bound semantics still fail on roughly half of applicable rows** (120 of 265, 45%), unchanged across all three model generations.
- EBM remains the weakest model on the highest-confidence tier** (A_STRICT_REAL R^2 = -3.93), consistent across v1/v3/v4.
- Pooled/overall statistics remain paper-dominated, not sample-representative.** One paper (<https://doi.org/10.1016/j.idairyj.2023.105596>, n=60, ~19% of the test set, MAE=75.3d) single-handedly drags every pooled number; per-tier and per-category breakdowns remain more trustworthy than the single pooled R^2 .

13. Recommendations

What v4 already achieved, and shouldn't be undone: the A_STRICT_REAL/feta fix and hard cheese's continued improvement are both real and should be preserved. The problem is not that v4 recalibrated — it's that the recalibration of semi-hard matrices (processed cheese, kashar, gouda) moved the label in the wrong direction while other matrices moved correctly.

Recommended, in order of expected payoff:

- Re-examine the specific v4 label-generation step for processed cheese, kashar, and gouda — these three matrices' labels moved substantially *up* from v3 (\$8.4), while the overall recalibration intent (based on the v3 $\rightarrow$ v4 fixes seen elsewhere) appears to have been to bring hard/A_STRICT_REAL matrices *down* toward real values. This looks like a sign-flipped or mis-scoped adjustment for this specific subset, not a fundamental data problem.
- Separately, address the pathogen-challenge blind spot directly: add indicator_type-aware short-shelf-life training examples for processed cheese (and any other matrix with C_SAFETY_EVENT_REAL coverage) so the model can learn that a deliberate high-inoculum

challenge study produces a very different outcome than a background-safety control at the same temperature. This is a data-coverage gap, not a calibration-scale problem, and predates v4.

3. Continue the Parmigiano-style approach for hard cheese: the label-to-live-output gap is narrowing but not closed; keep tracking it explicitly in the next iteration rather than assuming the label fix alone will finish the job.
4. Soft cheese continues to need no intervention ($R^2 = 0.34$).
5. Given the whack-a-mole pattern across v3→v4 (fixing hard/A_STRICT_REAL while breaking semi-hard), any future recalibration pass should re-run this full 319-case evaluation *before* declaring success, specifically checking every category and tier this report tracks, not just the one that motivated the change.

14. Appendix — full per-row results, all 319 rows, all 4 models (v4)

Sorted by cheese category, then confidence tier, then food matrix. "Err" = prediction – true value in days (positive = overprediction, negative = underprediction).

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0053	cheddar	hard	B_LOWER_BOUND_REAL	20.0	143.0	+123.0	146.7	+126.7	154.0	+134.0	155.8	+135.8
EXT_REAL_0054	cheddar	hard	B_LOWER_BOUND_REAL	30.0	142.9	+112.9	144.6	+114.6	153.9	+123.9	153.8	+123.8
EXT_REAL_0100	cheddar	hard	B_LOWER_BOUND_REAL	90.0	89.7	-0.3	83.1	-6.9	74.6	-15.4	85.9	-4.1
EXT_REAL_0101	cheddar	hard	B_LOWER_BOUND_REAL	90.0	57.3	-32.7	50.0	-40.0	45.9	-44.1	45.4	-44.6
EXT_REAL_0102	cheddar	hard	B_LOWER_BOUND_REAL	30.0	33.8	+3.8	32.3	+2.3	21.2	-8.8	0.0	-30.0
EXT_REAL_0103	cheddar	hard	B_LOWER_BOUND_REAL	90.0	88.7	-1.3	83.6	-6.4	74.3	-15.7	81.0	-9.0
EXT_REAL_0104	cheddar	hard	B_LOWER_BOUND_REAL	90.0	57.3	-32.7	50.2	-39.8	45.3	-44.7	40.5	-49.5
EXT_REAL_0105	cheddar	hard	B_LOWER_BOUND_REAL	30.0	36.7	+6.7	32.6	+2.6	25.7	-4.3	0.0	-30.0
EXT_REAL_0106	cheddar	hard	B_LOWER_BOUND_REAL	90.0	89.0	-1.0	80.1	-9.9	71.9	-18.1	83.4	-6.6
EXT_REAL_0107	cheddar	hard	B_LOWER_BOUND_REAL	90.0	56.5	-33.5	48.8	-41.2	43.1	-46.9	42.9	-47.1
EXT_REAL_0108	cheddar	hard	B_LOWER_BOUND_REAL	30.0	34.4	+4.4	30.7	+0.7	23.8	-6.2	0.0	-30.0
EXT_REAL_0109	cheddar	hard	B_LOWER_BOUND_REAL	90.0	89.2	-0.8	78.4	-11.6	71.0	-19.0	84.8	-5.2
EXT_REAL_0110	cheddar	hard	B_LOWER_BOUND_REAL	90.0	54.4	-35.6	47.4	-42.6	41.6	-48.4	44.3	-45.7
EXT_REAL_0111	cheddar	hard	B_LOWER_BOUND_REAL	30.0	31.7	+1.7	30.1	+0.1	17.9	-12.1	0.0	-30.0
EXT_REAL_0261	cheddar	hard	C_SAFETY_EVENT_REAL	7.0	26.6	+19.6	23.4	+16.4	12.6	+5.6	0.0	-7.0
EXT_REAL_0262	cheddar	hard	C_SAFETY_EVENT_REAL	7.0	29.4	+22.4	29.6	+22.6	23.2	+16.2	20.3	+13.3
EXT_REAL_0263	cheddar	hard	C_SAFETY_EVENT_REAL	15.0	81.2	+66.2	75.4	+60.4	68.5	+53.5	75.8	+60.8
EXT_REAL_0264	cheddar	hard	C_SAFETY_EVENT_REAL	15.0	92.5	+77.5	94.7	+79.7	93.4	+78.4	116.3	+101.3
EXT_REAL_0304	cheddar	hard	C_SAFETY_EVENT_REAL	15.0	26.9	+11.9	28.0	+13.0	15.2	+0.2	8.2	-6.8
EXT_REAL_0305	cheddar	hard	C_SAFETY_EVENT_REAL	15.0	26.9	+11.9	36.3	+21.3	24.3	+9.3	15.6	+0.6
EXT_REAL_0306	cheddar	hard	C_SAFETY_EVENT_REAL	15.0	26.9	+11.9	33.7	+18.7	22.4	+7.4	16.7	+1.7
EXT_REAL_0307	cheddar	hard	C_SAFETY_EVENT_REAL	15.0	26.9	+11.9	33.7	+18.7	22.4	+7.4	13.1	-1.9
EXT_REAL_0265	cheddar	hard	C_STUDY_HORIZON_REAL	15.0	86.7	+71.7	91.5	+76.5	86.6	+71.6	90.0	+75.0
EXT_REAL_0266	cheddar	hard	C_STUDY_HORIZON_REAL	15.0	97.1	+82.1	109.9	+94.9	106.5	+91.5	130.5	+115.5
EXT_REAL_0267	cheddar	hard	C_STUDY_HORIZON_REAL	120.0	107.0	-13.0	105.3	-14.7	105.4	-14.6	116.9	-3.1
EXT_REAL_0268	cheddar	hard	C_STUDY_HORIZON_REAL	120.0	98.0	-22.0	100.5	-19.5	102.9	-17.1	111.9	-8.1
EXT_REAL_0214	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	118.9	+28.9	126.0	+36.0	117.3	+27.3	102.7	+12.7
EXT_REAL_0215	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	120.5	+30.5	125.8	+35.8	121.0	+31.0	110.7	+20.7
EXT_REAL_0216	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	119.8	+29.8	127.2	+37.2	120.3	+30.3	108.9	+18.9
EXT_REAL_0217	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	127.8	+37.8	133.9	+43.9	127.3	+37.3	113.9	+23.9
EXT_REAL_0218	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	130.9	+40.9	129.7	+39.7	124.9	+34.9	121.9	+31.9
EXT_REAL_0219	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	130.6	+40.6	136.9	+46.9	130.7	+40.7	120.1	+30.1
EXT_REAL_0220	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	125.1	+35.1	130.8	+40.8	120.6	+30.6	107.9	+17.9
EXT_REAL_0221	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	126.2	+36.2	127.7	+37.7	123.3	+33.3	115.8	+25.8
EXT_REAL_0222	parmigiano reggiano	hard	C_STUDY_HORIZON_REAL	90.0	126.7	+36.7	131.9	+41.9	122.9	+32.9	114.1	+24.1
EXT_REAL_0260	gouda	semi_hard	B_LOWER_BOUND_REAL	56.0	112.1	+56.1	104.0	+48.0	97.7	+41.7	101.9	+45.9
EXT_REAL_0223	processed cheese	semi_hard	B_OVERALL_REAL	365.0	247.1	-117.9	239.3	-125.7	234.6	-130.4	266.8	-98.2
EXT_REAL_0308	colby	semi_hard	C_SAFETY_EVENT_REAL	15.0	30.9	+15.9	25.0	+10.0	3.2	-11.8	0.0	-15.0

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0309	colby	semi_hard	C_SAFETY_EVENT_REAL	15.0	30.9	+15.9	34.9	+19.9	23.1	+8.1	0.0	-15.0
EXT_REAL_0310	colby	semi_hard	C_SAFETY_EVENT_REAL	15.0	30.9	+15.9	32.2	+17.2	22.2	+7.2	0.0	-15.0
EXT_REAL_0311	colby	semi_hard	C_SAFETY_EVENT_REAL	15.0	30.9	+15.9	32.2	+17.2	22.2	+7.2	0.0	-15.0
EXT_REAL_0312	havarti	semi_hard	C_SAFETY_EVENT_REAL	15.0	31.9	+16.9	17.6	+2.6	1.4	-13.6	0.0	-15.0
EXT_REAL_0313	havarti	semi_hard	C_SAFETY_EVENT_REAL	15.0	32.1	+17.1	23.0	+8.0	18.4	+3.4	0.0	-15.0
EXT_REAL_0314	havarti	semi_hard	C_SAFETY_EVENT_REAL	15.0	32.1	+17.1	23.8	+8.8	18.2	+3.2	0.0	-15.0
EXT_REAL_0315	havarti	semi_hard	C_SAFETY_EVENT_REAL	15.0	32.1	+17.1	23.8	+8.8	18.2	+3.2	0.0	-15.0
EXT_REAL_0224	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	270.0	246.6	-23.4	239.3	-30.7	234.5	-35.5	266.9	-3.1
EXT_REAL_0225	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	258.1	+244.1	246.1	+232.1	241.7	+227.7	234.3	+220.3
EXT_REAL_0226	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	258.1	+244.1	257.0	+243.0	258.0	+244.0	241.7	+227.7
EXT_REAL_0227	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	258.1	+244.1	257.4	+243.4	258.1	+244.1	239.2	+225.2
EXT_REAL_0228	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	3.0	256.4	+253.4	239.3	+236.3	250.0	+247.0	238.7	+235.7
EXT_REAL_0229	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	3.0	258.0	+255.0	241.7	+238.7	250.8	+247.8	246.1	+243.1
EXT_REAL_0232	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	124.7	+110.7	102.7	+88.7	94.3	+80.3	47.8	+33.8
EXT_REAL_0233	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	124.8	+110.8	109.4	+95.4	108.1	+94.1	55.2	+41.2
EXT_REAL_0234	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	14.0	124.8	+110.8	109.8	+95.8	107.9	+93.9	52.7	+38.7
EXT_REAL_0235	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	3.0	126.0	+123.0	93.9	+90.9	103.3	+100.3	52.1	+49.1
EXT_REAL_0236	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	3.0	126.4	+123.4	92.7	+89.7	101.1	+98.1	59.5	+56.5
EXT_REAL_0300	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	4.0	64.9	+60.9	72.0	+68.0	57.0	+53.0	0.0	-4.0
EXT_REAL_0301	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	4.0	65.0	+61.0	77.8	+73.8	70.2	+66.2	0.0	-4.0
EXT_REAL_0302	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	4.0	65.0	+61.0	77.5	+73.5	70.4	+66.4	0.0	-4.0
EXT_REAL_0303	processed cheese	semi_hard	C_SAFETY_EVENT_REAL	4.0	65.0	+61.0	77.8	+73.8	70.2	+66.2	0.0	-4.0
EXT_REAL_0285	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	239.5	-125.5	237.4	-127.6	239.2	-125.8	251.1	-113.9
EXT_REAL_0286	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	237.6	-127.4	227.0	-138.0	233.4	-131.6	246.6	-118.4
EXT_REAL_0287	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	235.6	-129.4	236.1	-128.9	235.1	-129.9	251.1	-113.9
EXT_REAL_0288	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	380.1	+15.1	339.0	-26.0	331.0	-34.0	367.4	+2.4
EXT_REAL_0289	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	381.2	+16.2	331.1	-33.9	323.8	-41.2	362.9	-2.1
EXT_REAL_0290	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	382.6	+17.6	337.5	-27.5	327.2	-37.8	367.4	+2.4
EXT_REAL_0291	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	378.6	+13.6	339.0	-26.0	340.9	-24.1	377.9	+12.9
EXT_REAL_0292	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	379.9	+14.9	331.1	-33.9	332.5	-32.5	373.4	+8.4
EXT_REAL_0293	gouda	semi_hard	C_STUDY_HORIZON_REAL	365.0	380.6	+15.6	337.5	-27.5	338.7	-26.3	377.9	+12.9
EXT_REAL_0122	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	129.6	+84.6	129.2	+84.2	131.4	+86.4	145.3	+100.3
EXT_REAL_0123	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	130.1	+85.1	129.8	+84.8	133.3	+88.3	148.3	+103.3
EXT_REAL_0124	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	128.9	+83.9	127.7	+82.7	132.2	+87.2	142.0	+97.0
EXT_REAL_0125	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	138.0	+93.0	134.5	+89.5	138.7	+93.7	147.9	+102.9
EXT_REAL_0126	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	136.9	+91.9	134.8	+89.8	141.6	+96.6	139.9	+94.9
EXT_REAL_0127	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	155.1	+110.1	185.1	+140.1	176.4	+131.4	186.5	+141.5
EXT_REAL_0128	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	155.4	+110.4	191.7	+146.7	181.5	+136.5	189.5	+144.5
EXT_REAL_0129	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	155.0	+110.0	184.4	+139.4	176.7	+131.7	183.2	+138.2

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0130	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	163.4	+118.4	191.8	+146.8	181.6	+136.6	189.1	+144.1
EXT_REAL_0131	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	161.5	+116.5	195.8	+150.8	184.6	+139.6	181.1	+136.1
EXT_REAL_0132	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	153.1	+108.1	155.2	+110.2	150.3	+105.3	172.1	+127.1
EXT_REAL_0133	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	153.2	+108.2	160.3	+115.3	154.4	+109.4	175.2	+130.2
EXT_REAL_0134	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	152.5	+107.5	154.0	+109.0	150.3	+105.3	168.9	+123.9
EXT_REAL_0135	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	162.0	+117.0	161.2	+116.2	156.3	+111.3	174.8	+129.8
EXT_REAL_0136	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	160.1	+115.1	165.5	+120.5	159.5	+114.5	166.8	+121.8
EXT_REAL_0137	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	153.2	+108.2	178.3	+133.3	170.6	+125.6	182.3	+137.3
EXT_REAL_0138	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	153.3	+108.3	182.5	+137.5	174.5	+129.5	185.4	+140.4
EXT_REAL_0139	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	152.6	+107.6	177.4	+132.4	171.0	+126.0	179.0	+134.0
EXT_REAL_0140	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	162.2	+117.2	185.0	+140.0	176.7	+131.7	185.0	+140.0
EXT_REAL_0141	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	160.3	+115.3	188.2	+143.2	179.8	+134.8	177.0	+132.0
EXT_REAL_0142	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	70.7	+35.7	67.6	+32.6	68.5	+33.5	76.2	+41.2
EXT_REAL_0143	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	70.7	+35.7	68.3	+33.3	70.9	+35.9	79.2	+44.2
EXT_REAL_0144	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	70.7	+35.7	66.3	+31.3	69.4	+34.4	72.9	+37.9
EXT_REAL_0145	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	70.2	+35.2	71.3	+36.3	69.6	+34.6	78.8	+43.8
EXT_REAL_0146	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	71.3	+36.3	71.7	+36.7	70.7	+35.7	70.8	+35.8
EXT_REAL_0147	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.2	+45.2	102.4	+67.4	99.7	+64.7	117.4	+82.4
EXT_REAL_0148	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.3	+45.3	111.8	+76.8	105.4	+70.4	120.4	+85.4
EXT_REAL_0149	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.2	+45.2	103.1	+68.1	100.1	+65.1	114.1	+79.1
EXT_REAL_0150	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	79.9	+44.9	105.6	+70.6	101.8	+66.8	120.0	+85.0
EXT_REAL_0151	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.6	+45.6	107.3	+72.3	103.0	+68.0	112.0	+77.0
EXT_REAL_0152	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.7	+45.7	76.7	+41.7	80.1	+45.1	103.0	+68.0
EXT_REAL_0153	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.8	+45.8	84.9	+49.9	84.8	+49.8	106.1	+71.1
EXT_REAL_0154	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.7	+45.7	77.2	+42.2	80.1	+45.1	99.7	+64.7
EXT_REAL_0155	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.3	+45.3	80.0	+45.0	82.6	+47.6	105.7	+70.7
EXT_REAL_0156	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	81.0	+46.0	82.0	+47.0	83.5	+48.5	97.7	+62.7
EXT_REAL_0157	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	81.3	+46.3	98.0	+63.0	100.3	+65.3	113.2	+78.2
EXT_REAL_0158	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	81.3	+46.3	105.0	+70.0	104.8	+69.8	116.3	+81.3
EXT_REAL_0159	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	81.3	+46.3	98.6	+63.6	100.7	+65.7	109.9	+74.9
EXT_REAL_0160	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	80.8	+45.8	101.8	+66.8	103.1	+68.1	115.8	+80.8
EXT_REAL_0161	kashar	semi_hard	C_STUDY_HORIZON_REAL	35.0	81.5	+46.5	102.7	+67.7	104.0	+69.0	107.9	+72.9
EXT_REAL_0162	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	37.4	+22.4	53.7	+38.7	52.3	+37.3	32.1	+17.1
EXT_REAL_0163	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	37.5	+22.5	52.3	+37.3	54.1	+39.1	35.1	+20.1
EXT_REAL_0164	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	37.5	+22.5	50.0	+35.0	52.7	+37.7	28.8	+13.8
EXT_REAL_0165	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	37.5	+22.5	57.9	+42.9	54.1	+39.1	34.7	+19.7
EXT_REAL_0166	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	38.2	+23.2	54.6	+39.6	55.4	+40.4	26.7	+11.7
EXT_REAL_0167	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	41.6	+26.6	81.5	+66.5	80.1	+65.1	73.2	+58.2
EXT_REAL_0168	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.1	+27.1	88.7	+73.7	85.3	+70.3	76.3	+61.3
EXT_REAL_0169	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.0	+27.0	79.8	+64.8	80.0	+65.0	70.0	+55.0
EXT_REAL_0170	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	41.6	+26.6	82.6	+67.6	82.7	+67.7	75.9	+60.9
EXT_REAL_0171	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.3	+27.3	80.3	+65.3	84.2	+69.2	67.9	+52.9
EXT_REAL_0172	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	41.9	+26.9	56.5	+41.5	60.3	+45.3	58.9	+43.9
EXT_REAL_0173	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.3	+27.3	62.5	+47.5	64.9	+49.9	61.9	+46.9
EXT_REAL_0174	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.2	+27.2	54.7	+39.7	60.2	+45.2	55.6	+40.6
EXT_REAL_0175	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	41.9	+26.9	57.7	+42.7	63.6	+48.6	61.5	+46.5
EXT_REAL_0176	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.6	+27.6	55.8	+40.8	64.7	+49.7	53.5	+38.5

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0177	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.0	+27.0	77.6	+62.6	81.4	+66.4	69.1	+54.1
EXT_REAL_0178	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.4	+27.4	83.8	+68.8	85.4	+70.4	72.1	+57.1
EXT_REAL_0179	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.2	+27.2	77.1	+62.1	81.3	+66.3	65.8	+50.8
EXT_REAL_0180	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.0	+27.0	80.0	+65.0	85.0	+70.0	71.7	+56.7
EXT_REAL_0181	kashar	semi_hard	C_STUDY_HORIZON_REAL	15.0	42.7	+27.7	76.9	+61.9	86.0	+71.0	63.7	+48.7
EXT_REAL_0182	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	109.5	+79.5	93.3	+63.3	95.2	+65.2	100.9	+70.9
EXT_REAL_0183	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	133.8	+103.8	108.5	+78.5	106.8	+76.8	132.9	+102.9
EXT_REAL_0184	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	133.9	+103.9	108.7	+78.7	107.4	+77.4	135.8	+105.8
EXT_REAL_0185	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	135.5	+105.5	108.8	+78.8	107.9	+77.9	139.3	+109.3
EXT_REAL_0186	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	135.6	+105.6	108.8	+78.8	107.9	+77.9	136.9	+106.9
EXT_REAL_0187	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	109.5	+79.5	93.3	+63.3	94.3	+64.3	101.1	+71.1
EXT_REAL_0188	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	134.1	+104.1	108.1	+78.1	105.2	+75.2	127.5	+97.5
EXT_REAL_0189	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	133.8	+103.8	108.5	+78.5	105.9	+75.9	133.0	+103.0
EXT_REAL_0190	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	133.9	+103.9	108.7	+78.7	106.4	+76.4	135.9	+105.9
EXT_REAL_0191	kashar	semi_hard	C_STUDY_HORIZON_REAL	30.0	135.5	+105.5	108.8	+78.8	106.9	+76.9	139.4	+109.4
EXT_REAL_0192	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	167.2	+122.2	174.3	+129.3	180.9	+135.9	181.7	+136.7
EXT_REAL_0193	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	167.3	+122.3	174.7	+129.7	181.7	+136.7	184.2	+139.2
EXT_REAL_0194	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	168.0	+123.0	185.5	+140.5	184.3	+139.3	201.1	+156.1
EXT_REAL_0195	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	168.0	+123.0	186.4	+141.4	185.5	+140.5	201.7	+156.7
EXT_REAL_0196	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	168.0	+123.0	190.4	+145.4	190.3	+145.3	212.2	+167.2
EXT_REAL_0197	kashar	semi_hard	C_STUDY_HORIZON_REAL	45.0	168.0	+123.0	190.4	+145.4	190.3	+145.3	212.2	+167.2
EXT_REAL_0230	processed cheese	semi_hard	C_STUDY_HORIZON_REAL	84.0	257.5	+173.5	258.0	+174.0	258.1	+174.1	242.8	+158.8
EXT_REAL_0231	processed cheese	semi_hard	C_STUDY_HORIZON_REAL	84.0	278.6	+194.6	261.1	+177.1	280.7	+196.7	263.6	+179.6
EXT_REAL_0237	processed cheese	semi_hard	C_STUDY_HORIZON_REAL	84.0	124.4	+40.4	110.2	+26.2	107.9	+23.9	56.3	-27.7
EXT_REAL_0238	processed cheese	semi_hard	C_STUDY_HORIZON_REAL	84.0	126.9	+42.9	89.8	+5.8	118.7	+34.7	77.0	-7.0
EXT_REAL_0033	ricotta	soft	A_RANGE_REAL	3.0	7.9	+4.9	8.8	+5.8	9.4	+6.4	2.8	-0.2
EXT_REAL_0034	ricotta	soft	A_RANGE_REAL	3.0	6.8	+3.8	8.1	+5.1	8.9	+5.9	1.9	-1.1
EXT_REAL_0035	ricotta	soft	A_RANGE_REAL	3.0	7.9	+4.9	9.0	+6.0	9.5	+6.5	4.9	+1.9
EXT_REAL_0044	ricotta	soft	A_RANGE_REAL	10.5	7.3	-3.2	3.4	-7.1	2.5	-8.0	0.0	-10.5
EXT_REAL_0120	feta	soft	A_STRICT_REAL	10.0	18.1	+8.1	10.9	+0.9	13.8	+3.8	0.0	-10.0
EXT_REAL_0121	feta	soft	A_STRICT_REAL	17.0	18.9	+1.9	16.7	-0.3	19.3	+2.3	0.0	-17.0
EXT_REAL_0017	fresh cheese	soft	A_STRICT_REAL	9.0	9.9	+0.9	7.3	-1.7	10.2	+1.2	14.4	+5.4
EXT_REAL_0018	fresh cheese	soft	A_STRICT_REAL	9.0	9.9	+0.9	7.7	-1.3	10.6	+1.6	19.8	+10.8
EXT_REAL_0019	fresh cheese	soft	A_STRICT_REAL	9.0	12.4	+3.4	8.9	-0.1	12.5	+3.5	25.2	+16.2
EXT_REAL_0020	fresh cheese	soft	A_STRICT_REAL	9.0	10.5	+1.5	8.3	-0.7	11.2	+2.2	23.0	+14.0
EXT_REAL_0001	fresh mozzarella	soft	A_STRICT_REAL	4.0	6.1	+2.1	5.7	+1.7	3.6	-0.4	0.0	-4.0
EXT_REAL_0002	fresh mozzarella	soft	A_STRICT_REAL	4.0	7.1	+3.1	5.0	+1.0	4.6	+0.6	0.0	-4.0
EXT_REAL_0003	fresh mozzarella	soft	A_STRICT_REAL	4.0	6.1	+2.1	5.7	+1.7	3.6	-0.4	0.0	-4.0
EXT_REAL_0004	fresh mozzarella	soft	A_STRICT_REAL	4.0	6.9	+2.9	6.1	+2.1	5.9	+1.9	0.0	-4.0
EXT_REAL_0005	fresh mozzarella	soft	A_STRICT_REAL	8.0	6.8	-1.2	7.2	-0.8	4.7	-3.3	0.0	-8.0

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0006	fresh mozzarella	soft	A_STRICT_REAL	8.0	7.9	-0.1	6.5	-1.5	5.8	-2.2	0.0	-8.0
EXT_REAL_0007	fresh mozzarella	soft	A_STRICT_REAL	8.0	6.8	-1.2	7.2	-0.8	4.7	-3.3	0.0	-8.0
EXT_REAL_0008	fresh mozzarella	soft	A_STRICT_REAL	8.0	7.7	-0.3	9.2	+1.2	8.4	+0.4	0.0	-8.0
EXT_REAL_0009	fresh mozzarella	soft	A_STRICT_REAL	8.0	6.6	-1.4	7.6	-0.4	6.5	-1.5	0.0	-8.0
EXT_REAL_0010	fresh mozzarella	soft	A_STRICT_REAL	8.0	7.8	-0.2	6.0	-2.0	6.6	-1.4	0.0	-8.0
EXT_REAL_0011	fresh mozzarella	soft	A_STRICT_REAL	8.0	6.6	-1.4	7.6	-0.4	6.5	-1.5	0.0	-8.0
EXT_REAL_0012	fresh mozzarella	soft	A_STRICT_REAL	8.0	7.6	-0.4	8.4	+0.4	9.2	+1.2	4.6	-3.4
EXT_REAL_0013	fresh mozzarella	soft	A_STRICT_REAL	8.0	6.7	-1.3	9.5	+1.5	7.9	-0.1	5.8	-2.2
EXT_REAL_0014	fresh mozzarella	soft	A_STRICT_REAL	8.0	7.9	-0.1	8.0	-0.0	8.0	+0.0	0.0	-8.0
EXT_REAL_0015	fresh mozzarella	soft	A_STRICT_REAL	8.0	6.8	-1.2	9.5	+1.5	7.9	-0.1	6.8	-1.2
EXT_REAL_0016	fresh mozzarella	soft	A_STRICT_REAL	8.0	7.7	-0.3	11.5	+3.5	11.4	+3.4	11.7	+3.7
EXT_REAL_0056	paneer	soft	A_STRICT_REAL	7.0	12.8	+5.8	8.7	+1.7	7.0	-0.0	1.1	-5.9
EXT_REAL_0057	paneer	soft	A_STRICT_REAL	21.0	15.7	-5.3	11.0	-10.0	10.9	-10.1	0.0	-21.0
EXT_REAL_0039	ricotta	soft	A_STRICT_REAL	15.0	8.8	-6.2	9.9	-5.1	9.5	-5.5	18.6	+3.6
EXT_REAL_0040	ricotta	soft	A_STRICT_REAL	15.0	8.9	-6.1	9.7	-5.3	9.1	-5.9	17.7	+2.7
EXT_REAL_0041	ricotta	soft	A_STRICT_REAL	15.0	8.8	-6.2	10.6	-4.4	9.8	-5.2	20.8	+5.8
EXT_REAL_0079	cottage cheese	soft	B_LOWER_BOUND_REAL	63.0	12.1	-50.9	8.2	-54.8	7.5	-55.5	0.0	-63.0
EXT_REAL_0080	cottage cheese	soft	B_LOWER_BOUND_REAL	63.0	12.3	-50.7	13.8	-49.2	16.7	-46.3	15.3	-47.7
EXT_REAL_0081	cottage cheese	soft	B_LOWER_BOUND_REAL	63.0	11.0	-52.0	5.8	-57.2	3.5	-59.5	0.0	-63.0
EXT_REAL_0082	cottage cheese	soft	B_LOWER_BOUND_REAL	63.0	11.1	-51.9	9.9	-53.1	8.1	-54.9	2.3	-60.7
EXT_REAL_0083	cottage cheese	soft	B_LOWER_BOUND_REAL	63.0	11.8	-51.2	11.6	-51.4	15.4	-47.6	10.6	-52.4
EXT_REAL_0084	cottage cheese	soft	B_LOWER_BOUND_REAL	63.0	10.4	-52.6	8.3	-54.7	5.3	-57.7	0.0	-63.0
EXT_REAL_0269	feta	soft	B_LOWER_BOUND_REAL	180.0	57.4	-122.6	55.5	-124.5	61.1	-118.9	53.6	-126.4
EXT_REAL_0270	feta	soft	B_LOWER_BOUND_REAL	180.0	58.5	-121.5	52.1	-127.9	53.9	-126.1	59.5	-120.5
EXT_REAL_0271	feta	soft	B_LOWER_BOUND_REAL	180.0	59.0	-121.0	53.9	-126.1	53.3	-126.7	64.0	-116.0
EXT_REAL_0272	feta	soft	B_LOWER_BOUND_REAL	180.0	59.5	-120.5	53.9	-126.1	54.2	-125.8	62.2	-117.8
EXT_REAL_0273	feta	soft	B_LOWER_BOUND_REAL	180.0	151.7	-28.3	134.9	-45.1	157.3	-22.7	75.4	-104.6
EXT_REAL_0274	feta	soft	B_LOWER_BOUND_REAL	180.0	152.0	-28.0	138.7	-41.3	140.1	-39.9	81.3	-98.7
EXT_REAL_0275	feta	soft	B_LOWER_BOUND_REAL	180.0	152.5	-27.5	142.2	-37.8	142.9	-37.1	85.8	-94.2
EXT_REAL_0276	feta	soft	B_LOWER_BOUND_REAL	180.0	154.5	-25.5	141.2	-38.8	142.1	-37.9	84.0	-96.0
EXT_REAL_0277	feta	soft	B_LOWER_BOUND_REAL	180.0	31.2	-148.8	30.7	-149.3	37.3	-142.7	0.0	-180.0
EXT_REAL_0278	feta	soft	B_LOWER_BOUND_REAL	180.0	32.8	-147.2	23.9	-156.1	27.3	-152.7	0.0	-180.0
EXT_REAL_0279	feta	soft	B_LOWER_BOUND_REAL	180.0	33.0	-147.0	23.8	-156.2	25.6	-154.4	0.0	-180.0
EXT_REAL_0280	feta	soft	B_LOWER_BOUND_REAL	180.0	33.1	-146.9	25.2	-154.8	29.2	-150.8	0.0	-180.0
EXT_REAL_0281	feta	soft	B_LOWER_BOUND_REAL	180.0	72.2	-107.8	50.8	-129.2	78.5	-101.5	0.0	-180.0
EXT_REAL_0282	feta	soft	B_LOWER_BOUND_REAL	180.0	74.1	-105.9	45.6	-134.4	54.7	-125.3	0.0	-180.0
EXT_REAL_0283	feta	soft	B_LOWER_BOUND_REAL	180.0	74.6	-105.4	48.2	-131.8	58.3	-121.7	0.2	-179.8
EXT_REAL_0284	feta	soft	B_LOWER_BOUND_REAL	180.0	75.2	-104.8	47.6	-132.4	58.2	-121.8	0.0	-180.0

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0021	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	16.4	-1.6	12.7	-5.3	18.2	+0.2	29.3	+11.3
EXT_REAL_0022	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	16.0	-2.0	13.3	-4.7	18.7	+0.7	34.7	+16.7
EXT_REAL_0023	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	16.6	-1.4	14.4	-3.6	19.8	+1.8	40.1	+22.1
EXT_REAL_0024	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	16.4	-1.6	13.6	-4.4	18.9	+0.9	38.0	+20.0
EXT_REAL_0025	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.7	-0.3	14.9	-3.1	20.4	+2.4	54.8	+36.8
EXT_REAL_0026	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.3	-0.7	15.4	-2.6	20.8	+2.8	60.3	+42.3
EXT_REAL_0027	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.8	-0.2	16.6	-1.4	21.5	+3.5	65.7	+47.7
EXT_REAL_0028	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.8	-0.2	15.8	-2.2	21.2	+3.2	63.5	+45.5
EXT_REAL_0029	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.6	-0.4	14.9	-3.1	20.4	+2.4	34.4	+16.4
EXT_REAL_0030	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.1	-0.9	15.4	-2.6	20.8	+2.8	39.9	+21.9
EXT_REAL_0031	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.8	-0.2	16.6	-1.4	21.5	+3.5	45.3	+27.3
EXT_REAL_0032	fresh cheese	soft	B_LOWER_BOUND_REAL	18.0	17.6	-0.4	15.8	-2.2	21.2	+3.2	43.1	+25.1
EXT_REAL_0058	fresh cheese	soft	B_LOWER_BOUND_REAL	14.0	13.2	-0.8	13.2	-0.8	15.7	+1.7	31.3	+17.3
EXT_REAL_0042	minas fresh cheese	soft	B_LOWER_BOUND_REAL	28.0	16.0	-12.0	8.4	-19.6	10.6	-17.4	0.0	-28.0
EXT_REAL_0043	minas fresh cheese	soft	B_LOWER_BOUND_REAL	28.0	15.9	-12.1	9.3	-18.7	10.9	-17.1	0.0	-28.0
EXT_REAL_0045	ricotta	soft	B_LOWER_BOUND_REAL	56.0	8.0	-48.0	6.0	-50.0	5.6	-50.4	0.0	-56.0
EXT_REAL_0055	soft white cheese	soft	B_LOWER_BOUND_REAL	30.0	17.2	-12.8	11.5	-18.5	13.4	-16.6	30.4	+0.4
EXT_REAL_0071	cottage cheese	soft	B_OVERALL_REAL	19.0	14.1	-4.9	14.6	-4.4	13.9	-5.1	7.1	-11.9
EXT_REAL_0072	cottage cheese	soft	B_OVERALL_REAL	19.0	17.3	-1.7	13.9	-5.1	13.9	-5.1	6.3	-12.7
EXT_REAL_0073	cottage cheese	soft	B_OVERALL_REAL	19.0	14.0	-5.0	14.3	-4.7	13.0	-6.0	8.9	-10.1
EXT_REAL_0074	cottage cheese	soft	B_OVERALL_REAL	6.5	12.5	+6.0	12.0	+5.5	9.5	+3.0	0.0	-6.5
EXT_REAL_0075	cottage cheese	soft	B_OVERALL_REAL	6.5	13.4	+6.9	11.8	+5.3	9.8	+3.3	0.0	-6.5
EXT_REAL_0076	cottage cheese	soft	B_OVERALL_REAL	6.5	12.4	+5.9	11.9	+5.4	5.5	-1.0	0.0	-6.5
EXT_REAL_0059	paneer	soft	B_OVERALL_REAL	9.0	10.5	+1.5	9.7	+0.7	6.9	-2.1	11.9	+2.9
EXT_REAL_0060	paneer	soft	B_OVERALL_REAL	9.0	10.2	+1.2	9.6	+0.6	7.0	-2.0	11.0	+2.0
EXT_REAL_0061	paneer	soft	B_OVERALL_REAL	9.0	9.9	+0.9	9.3	+0.3	6.3	-2.7	8.5	-0.5
EXT_REAL_0062	paneer	soft	B_OVERALL_REAL	9.0	10.5	+1.5	10.4	+1.4	2.2	-6.8	13.7	+4.7
EXT_REAL_0063	paneer	soft	B_OVERALL_REAL	9.0	10.7	+1.7	10.6	+1.6	6.8	-2.2	5.7	-3.3
EXT_REAL_0064	paneer	soft	B_OVERALL_REAL	9.0	10.2	+1.2	9.7	+0.7	6.5	-2.5	14.1	+5.1
EXT_REAL_0065	paneer	soft	B_OVERALL_REAL	15.0	12.5	-2.5	13.8	-1.2	10.2	-4.8	13.6	-1.4
EXT_REAL_0066	paneer	soft	B_OVERALL_REAL	15.0	12.4	-2.6	13.3	-1.7	10.2	-4.8	12.8	-2.2
EXT_REAL_0067	paneer	soft	B_OVERALL_REAL	15.0	11.9	-3.1	12.6	-2.4	9.4	-5.6	10.2	-4.8
EXT_REAL_0068	paneer	soft	B_OVERALL_REAL	15.0	12.4	-2.6	11.9	-3.1	3.5	-11.5	15.4	+0.4
EXT_REAL_0069	paneer	soft	B_OVERALL_REAL	15.0	12.7	-2.3	13.5	-1.5	8.9	-6.1	7.4	-7.6
EXT_REAL_0070	paneer	soft	B_OVERALL_REAL	15.0	12.0	-3.0	12.1	-2.9	8.8	-6.2	15.8	+0.8

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0036	ricotta	soft	B_OVERALL_REAL	7.5	8.2	+0.7	8.2	+0.7	8.4	+0.9	13.5	+6.0
EXT_REAL_0037	ricotta	soft	B_OVERALL_REAL	7.5	8.2	+0.7	8.0	+0.5	8.2	+0.7	12.6	+5.1
EXT_REAL_0038	ricotta	soft	B_OVERALL_REAL	7.5	8.2	+0.7	8.8	+1.3	8.8	+1.3	15.7	+8.2
EXT_REAL_0204	white cheese	soft	B_OVERALL_REAL	45.0	21.9	-23.1	15.8	-29.2	24.7	-20.3	48.9	+3.9
EXT_REAL_0077	cottage cheese	soft	C_SAFETY_EVENT_REAL	28.0	14.4	-13.6	6.8	-21.2	7.7	-20.3	0.0	-28.0
EXT_REAL_0078	cottage cheese	soft	C_SAFETY_EVENT_REAL	7.0	13.1	+6.1	4.8	-2.2	2.9	-4.1	0.0	-7.0
EXT_REAL_0112	feta	soft	C_SAFETY_EVENT_REAL	28.0	29.6	+1.6	15.1	-12.9	25.7	-2.3	22.9	-5.1
EXT_REAL_0113	feta	soft	C_SAFETY_EVENT_REAL	17.0	32.1	+15.1	22.2	+5.2	29.6	+12.6	38.3	+21.3
EXT_REAL_0114	feta	soft	C_SAFETY_EVENT_REAL	28.0	17.4	-10.6	6.7	-21.3	11.3	-16.7	0.0	-28.0
EXT_REAL_0115	feta	soft	C_SAFETY_EVENT_REAL	10.0	18.2	+8.2	13.6	+3.6	15.9	+5.9	0.0	-10.0
EXT_REAL_0116	feta	soft	C_SAFETY_EVENT_REAL	25.0	41.6	+16.6	45.5	+20.5	42.1	+17.1	33.6	+8.6
EXT_REAL_0117	feta	soft	C_SAFETY_EVENT_REAL	25.0	52.1	+27.1	53.7	+28.7	52.8	+27.8	49.0	+24.0
EXT_REAL_0118	feta	soft	C_SAFETY_EVENT_REAL	44.0	22.4	-21.6	21.0	-23.0	17.7	-26.3	0.0	-44.0
EXT_REAL_0119	feta	soft	C_SAFETY_EVENT_REAL	25.0	24.5	-0.5	27.2	+2.2	23.0	-2.0	0.0	-25.0
EXT_REAL_0316	fresh mozzarella	soft	C_SAFETY_EVENT_REAL	15.0	6.0	-9.0	0.0	-15.0	0.0	-15.0	0.0	-15.0
EXT_REAL_0317	fresh mozzarella	soft	C_SAFETY_EVENT_REAL	15.0	6.0	-9.0	0.0	-15.0	0.0	-15.0	0.0	-15.0
EXT_REAL_0318	fresh mozzarella	soft	C_SAFETY_EVENT_REAL	15.0	6.0	-9.0	0.0	-15.0	0.0	-15.0	0.0	-15.0
EXT_REAL_0319	fresh mozzarella	soft	C_SAFETY_EVENT_REAL	15.0	6.0	-9.0	0.0	-15.0	0.0	-15.0	0.0	-15.0
EXT_REAL_0046	queso fresco	soft	C_SAFETY_EVENT_REAL	7.0	5.8	-1.2	2.1	-4.9	1.4	-5.6	0.0	-7.0
EXT_REAL_0047	queso fresco	soft	C_SAFETY_EVENT_REAL	7.0	10.7	+3.7	4.7	-2.3	3.6	-3.4	0.0	-7.0
EXT_REAL_0048	queso fresco	soft	C_SAFETY_EVENT_REAL	7.0	8.5	+1.5	5.2	-1.8	3.8	-3.2	0.0	-7.0
EXT_REAL_0049	queso fresco	soft	C_SAFETY_EVENT_REAL	21.0	11.7	-9.3	5.9	-15.1	5.6	-15.4	0.0	-21.0
EXT_REAL_0050	queso fresco	soft	C_SAFETY_EVENT_REAL	21.0	12.2	-8.8	6.4	-14.6	6.2	-14.8	5.5	-15.5
EXT_REAL_0051	queso fresco	soft	C_SAFETY_EVENT_REAL	28.0	10.8	-17.2	5.9	-22.1	5.8	-22.2	0.0	-28.0
EXT_REAL_0052	queso fresco	soft	C_SAFETY_EVENT_REAL	35.0	10.8	-24.2	5.9	-29.1	5.8	-29.2	0.0	-35.0
EXT_REAL_0085	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	16.8	-11.2	18.3	-9.7	18.2	-9.8	4.6	-23.4
EXT_REAL_0086	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	19.0	-9.0	18.5	-9.5	18.8	-9.2	0.0	-28.0
EXT_REAL_0087	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	18.7	-9.3	18.8	-9.2	19.0	-9.0	2.5	-25.5
EXT_REAL_0088	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	15.7	-12.3	18.0	-10.0	18.4	-9.6	4.7	-23.3
EXT_REAL_0089	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	17.0	-11.0	18.3	-9.7	18.2	-9.8	21.6	-6.4
EXT_REAL_0090	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	19.2	-8.8	18.5	-9.5	18.8	-9.2	16.7	-11.3
EXT_REAL_0091	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	18.9	-9.1	18.8	-9.2	19.0	-9.0	19.4	-8.6
EXT_REAL_0092	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	15.9	-12.1	18.0	-10.0	18.4	-9.6	21.6	-6.4
EXT_REAL_0093	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	14.1	-13.9	16.0	-12.0	14.8	-13.2	10.5	-17.5

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0094	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	14.4	-13.6	16.0	-12.0	15.0	-13.0	5.6	-22.4
EXT_REAL_0095	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	13.9	-14.1	16.3	-11.7	15.2	-12.8	8.3	-19.7
EXT_REAL_0096	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	13.5	-14.5	15.6	-12.4	14.9	-13.1	10.5	-17.5
EXT_REAL_0097	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	14.5	-13.5	13.8	-14.2	13.3	-14.7	9.3	-18.7
EXT_REAL_0098	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	14.4	-13.6	13.8	-14.2	13.4	-14.6	4.4	-23.6
EXT_REAL_0099	cottage cheese	soft	C_STUDY_HORIZON_REAL	28.0	13.9	-14.1	13.6	-14.4	13.6	-14.4	9.3	-18.7
EXT_REAL_0257	cottage cheese	soft	C_STUDY_HORIZON_REAL	14.0	13.5	-0.5	10.2	-3.8	11.3	-2.7	0.0	-14.0
EXT_REAL_0258	cottage cheese	soft	C_STUDY_HORIZON_REAL	14.0	11.4	-2.6	8.0	-6.0	6.3	-7.7	0.0	-14.0
EXT_REAL_0259	cottage cheese	soft	C_STUDY_HORIZON_REAL	14.0	11.0	-3.0	6.1	-7.9	3.6	-10.4	0.0	-14.0
EXT_REAL_0239	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	24.4	-10.6	26.7	-8.3	39.2	+4.2	43.7	+8.7
EXT_REAL_0240	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	21.2	-13.8	47.9	+12.9	49.0	+14.0	48.7	+13.7
EXT_REAL_0241	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	23.5	-11.5	29.8	-5.2	24.9	-10.1	29.0	-6.0
EXT_REAL_0242	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	23.2	-11.8	29.8	-5.2	23.0	-12.0	28.1	-6.9
EXT_REAL_0243	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	23.2	-11.8	30.6	-4.4	22.1	-12.9	30.7	-4.3
EXT_REAL_0244	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	20.9	-14.1	29.4	-5.6	20.2	-14.8	31.2	-3.8
EXT_REAL_0245	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	28.4	-6.6	33.9	-1.1	47.0	+12.0	56.5	+21.5
EXT_REAL_0246	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	24.9	-10.1	59.4	+24.4	73.5	+38.5	61.5	+26.5
EXT_REAL_0247	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	27.6	-7.4	34.3	-0.7	29.6	-5.4	41.8	+6.8
EXT_REAL_0248	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	27.4	-7.6	34.4	-0.6	27.7	-7.3	40.9	+5.9
EXT_REAL_0249	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	27.5	-7.5	34.9	-0.1	29.2	-5.8	43.5	+8.5
EXT_REAL_0250	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	24.6	-10.4	33.9	-1.1	27.0	-8.0	44.0	+9.0
EXT_REAL_0251	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	28.2	-6.8	33.9	-1.1	47.0	+12.0	53.8	+18.8
EXT_REAL_0252	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	24.8	-10.2	59.4	+24.4	73.0	+38.0	58.7	+23.7
EXT_REAL_0253	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	27.5	-7.5	34.3	-0.7	29.6	-5.4	39.0	+4.0
EXT_REAL_0254	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	27.3	-7.7	34.4	-0.6	27.8	-7.2	38.2	+3.2
EXT_REAL_0255	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	27.3	-7.7	34.9	-0.1	29.2	-5.8	40.8	+5.8
EXT_REAL_0256	cream cheese	soft	C_STUDY_HORIZON_REAL	35.0	24.4	-10.6	33.9	-1.1	27.0	-8.0	41.2	+6.2
EXT_REAL_0294	queso fresco	soft	C_STUDY_HORIZON_REAL	28.0	8.0	-20.0	6.9	-21.1	7.4	-20.6	0.0	-28.0
EXT_REAL_0295	queso fresco	soft	C_STUDY_HORIZON_REAL	28.0	10.6	-17.4	10.2	-17.8	9.5	-18.5	0.0	-28.0
EXT_REAL_0296	queso fresco	soft	C_STUDY_HORIZON_REAL	28.0	6.4	-21.6	5.0	-23.0	3.0	-25.0	0.0	-28.0
EXT_REAL_0297	queso fresco	soft	C_STUDY_HORIZON_REAL	28.0	8.1	-19.9	8.6	-19.4	6.2	-21.8	0.0	-28.0

ROW ID	FOOD MATRIX	CATEGORY	TIER	TRUE (D)	RF	ERR	LIGHTGBM	ERR	XGBOOST	ERR	EBM	ERR
EXT_REAL_0298	queso fresco	soft	C_STUDY_HORIZON_REAL	28.0	5.7	-22.3	3.0	-25.0	1.1	-26.9	0.0	-28.0
EXT_REAL_0299	queso fresco	soft	C_STUDY_HORIZON_REAL	28.0	7.3	-20.7	6.4	-21.6	4.3	-23.7	0.0	-28.0
EXT_REAL_0205	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	6.3	-21.7	4.9	-23.1	5.6	-22.4	0.0	-28.0
EXT_REAL_0206	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	7.9	-20.1	8.3	-19.7	7.6	-20.4	0.6	-27.4
EXT_REAL_0207	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	8.0	-20.0	8.6	-19.4	8.3	-19.7	0.0	-28.0
EXT_REAL_0208	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	7.7	-20.3	6.3	-21.7	7.4	-20.6	0.0	-28.0
EXT_REAL_0209	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	9.3	-18.7	8.6	-19.4	9.0	-19.0	0.0	-28.0
EXT_REAL_0210	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	9.3	-18.7	9.0	-19.0	9.7	-18.3	0.0	-28.0
EXT_REAL_0211	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	8.1	-19.9	7.3	-20.7	7.3	-20.7	0.0	-28.0
EXT_REAL_0212	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	9.6	-18.4	9.2	-18.8	8.8	-19.2	3.7	-24.3
EXT_REAL_0213	ricotta	soft	C_STUDY_HORIZON_REAL	28.0	9.6	-18.4	10.1	-17.9	9.4	-18.6	0.0	-28.0
EXT_REAL_0198	white cheese	soft	C_STUDY_HORIZON_REAL	45.0	20.0	-25.0	29.1	-15.9	25.6	-19.4	38.3	-6.7
EXT_REAL_0199	white cheese	soft	C_STUDY_HORIZON_REAL	45.0	20.2	-24.8	16.7	-28.3	23.0	-22.0	39.0	-6.0
EXT_REAL_0200	white cheese	soft	C_STUDY_HORIZON_REAL	45.0	20.1	-24.9	29.0	-16.0	26.0	-19.0	40.2	-4.8
EXT_REAL_0201	white cheese	soft	C_STUDY_HORIZON_REAL	45.0	20.3	-24.7	16.6	-28.4	23.0	-22.0	40.9	-4.1
EXT_REAL_0202	white cheese	soft	C_STUDY_HORIZON_REAL	45.0	20.1	-24.9	29.0	-16.0	26.0	-19.0	41.2	-3.8
EXT_REAL_0203	white cheese	soft	C_STUDY_HORIZON_REAL	45.0	20.3	-24.7	16.6	-28.4	23.0	-22.0	41.8	-3.2

Report generated from stress_test_319_predictions.csv, produced by evaluate_319_stress_test.py against the live v4 production models (artifacts/, retrained from scratch on CHEESE_SHELF_LIFE_TARGETED_V4_READY_TO_TRAIN.xlsx). v1 and v3 comparison figures are read from model_versions/v1_restored_baseline_2026-08-18_141539/stress_test_319_predictions.csv and model_versions/v3_recalibrated_2026-08-19_124027/stress_test_319_predictions.csv respectively, both backups taken immediately before their successor retrain. No retraining occurred in the course of generating this evaluation or report.